RIVER ROUTING

ERT 474/574

Open-Source Hydro Data Analytics

Nov 2nd 2026

University at Buffalo The State University of New York





Why do we need a routing model??

What is the output from a hydrologic model?

Water balance equation

$$P = ET + R + \Delta S$$

Input

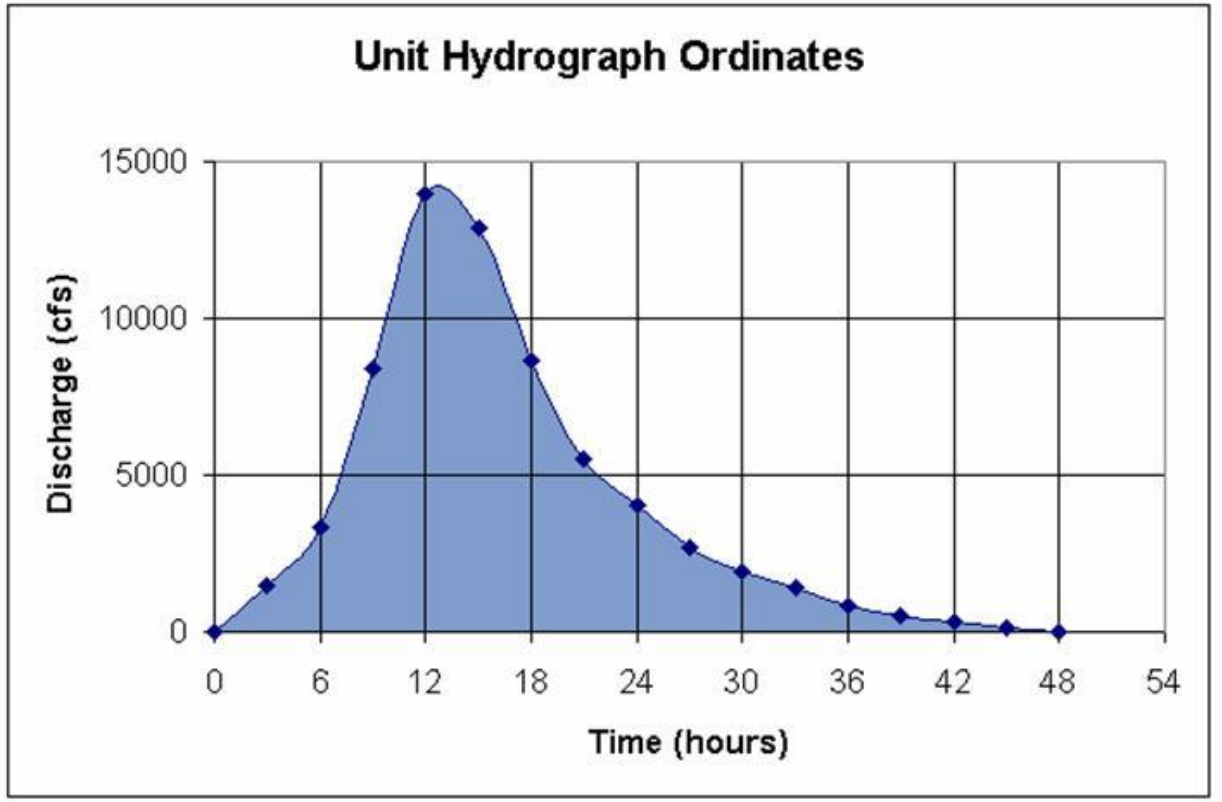
Output

We have already calculated the runoff in the hydrologic model, but measuring it is very challenging!

As runoff eventually will contribute to the streamflow in the river, we will need to take a step further and leverage a ***Routing Model*** to convert runoff into streamflow so that we can evaluate the accuracy of the numerical model.

Routing Scheme #1: Impulse Response Function (IRF)

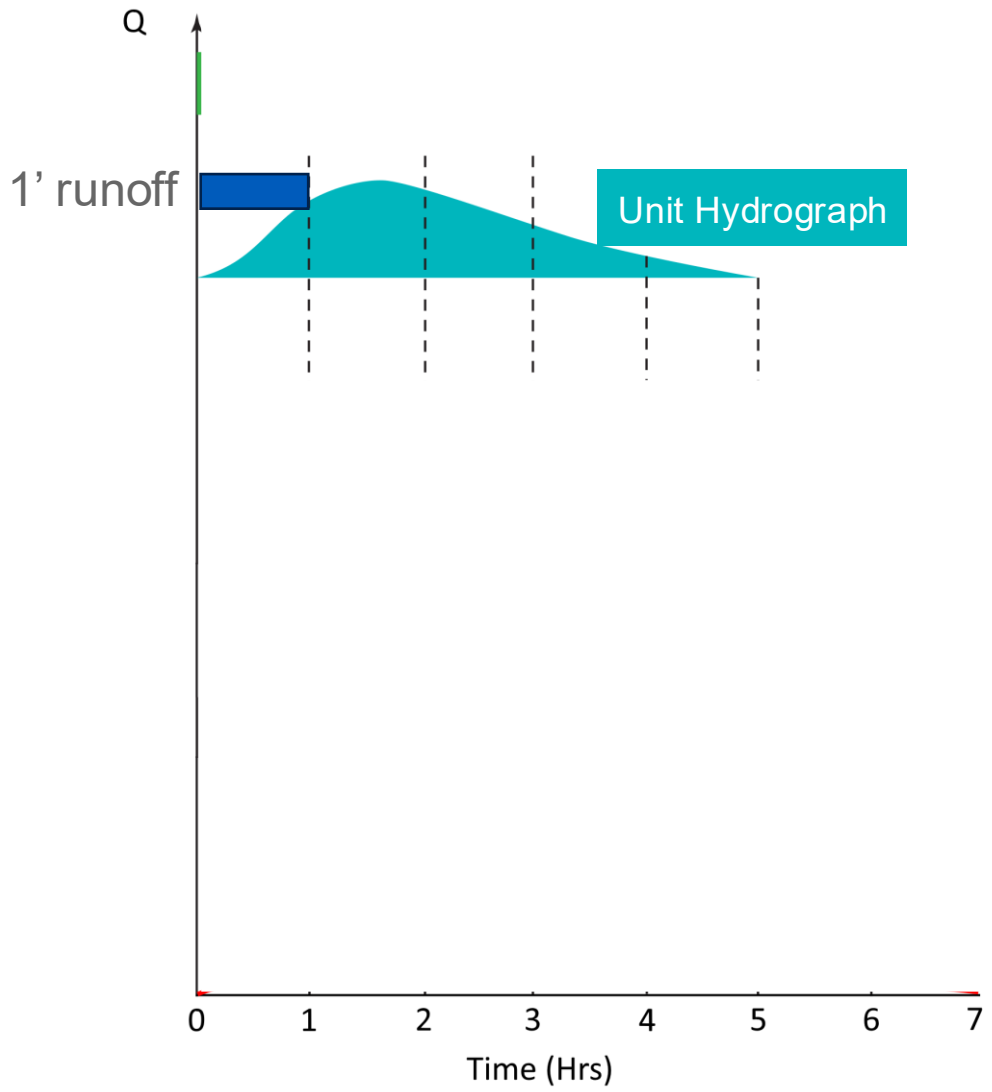
- IFR is used by the ***Lohmann*** Routing Scheme
  - IRF is called as the Unit Hydrograph (UH) by hydrologists
- What is a unit hydrograph?
  - A unit hydrograph represents the hydrograph from direct or surface runoff from one inch of runoff over the entire basin in a given unit of time.



Unit Hydrograph for Runoff

Unit hydrograph theory converts runoff values for a basin into flow or discharge at the basin outlet.

This method is used in the default routing model for VIC, called ***RVIC***.

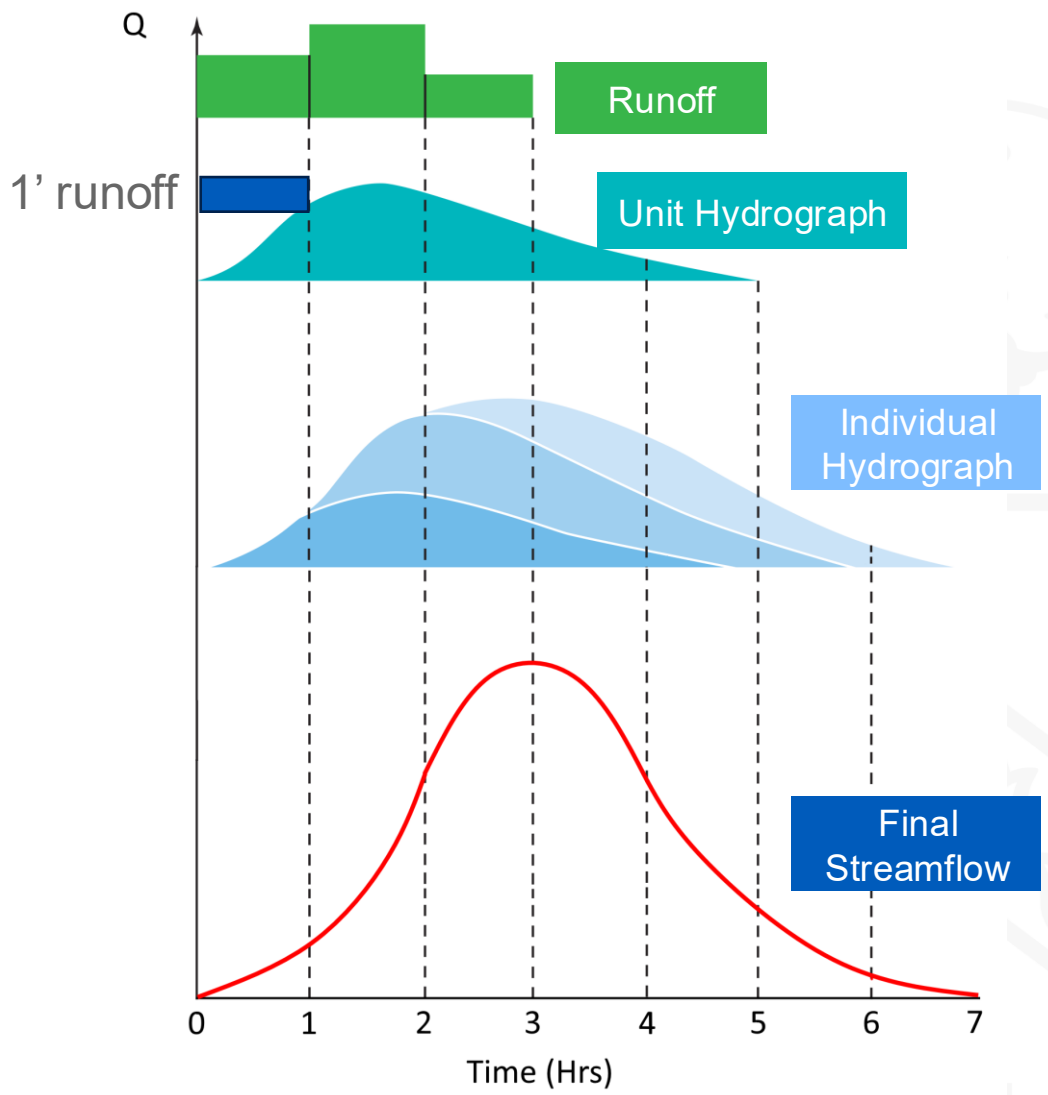


https://training.weather.gov/nwstc/Hydrology/HYDRO/BHModule/BH-unit5.HTML

Unit Hydrograph for Runoff

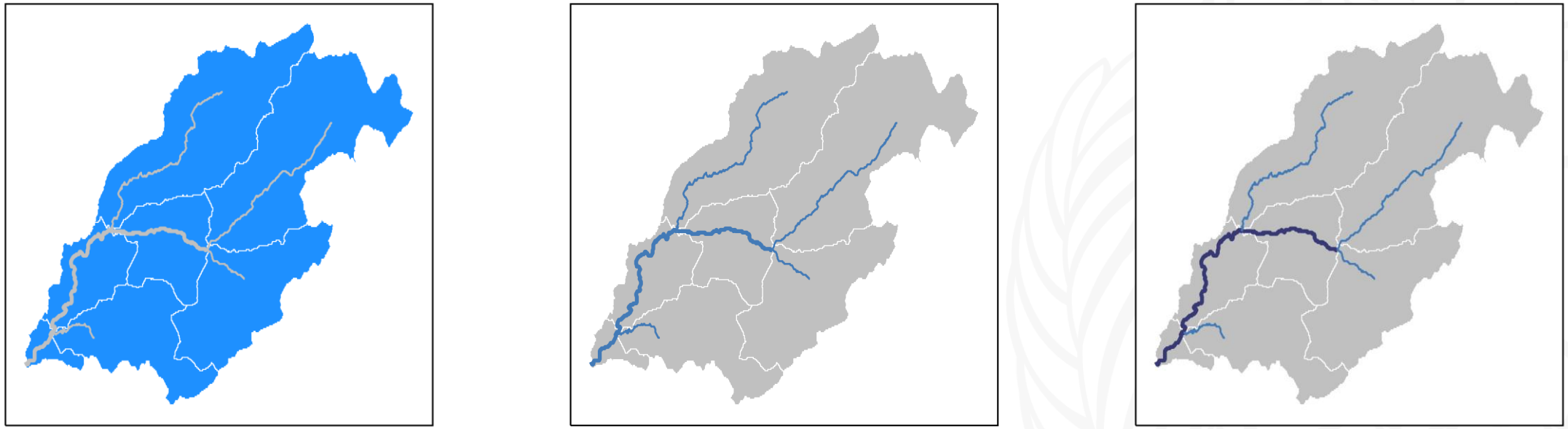
Unit hydrograph theory converts runoff values for a basin into flow or discharge at the basin outlet.

This method is used in the default routing model for VIC, called ***RVIC***.



https://training.weather.gov/nwstc/Hydrology/HYDRO/BHModule/BH-unit5.HTML

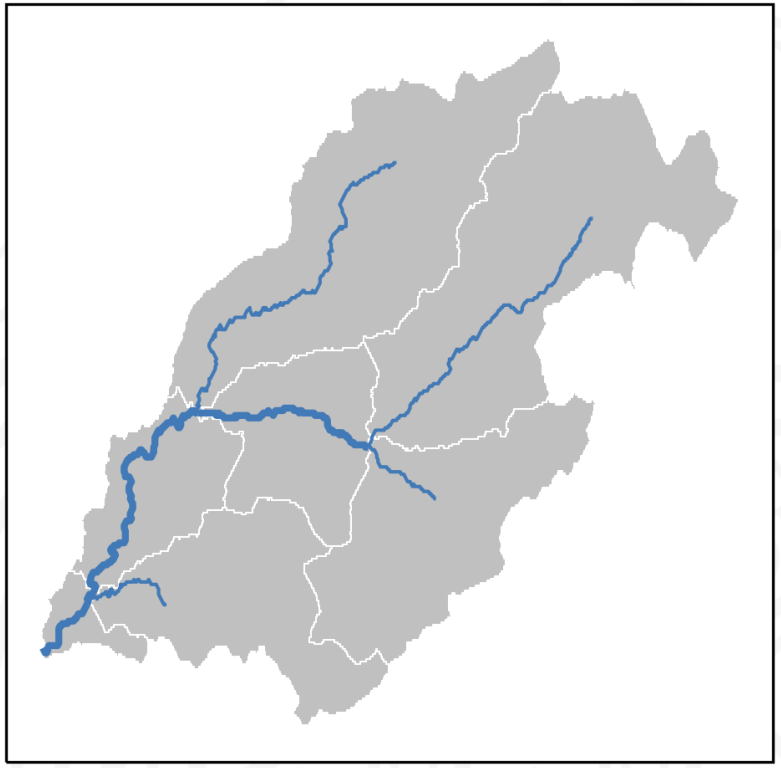
RVIC model separate the routing into two-steps



Hillslope routing is performed with a gamma-distribution-based unit-hydrograph to transport runoff from a hillslope to a catchment outlet

What are the assumptions for IRF?

- **Time-Invariant Impulse Response Functions (IRFs)**
  The routing behavior from each grid cell to a downstream point is characterized by an IRF, which is assumed to be linear and time-invariant. The response to runoff inputs does not change over time, which can be precomputed.
- **Neglect of Nonlinear Channel Processes**
  The model does not account for complex channel hydraulics, backwater effects, or floodplain interactions. It is best suited for large-scale applications where such details are less critical.



Each pair of the upstream and downstream HRUs has one unit hydrograph

Routing Scheme #2: Kinematic wave tracking (KWT)

- **KWT method** computes a wave speed or a celerity for the runoff (or discharge) that enters an individual stream segment from the corresponding HRU at each time step using kinematic approximation
  - The runoff, represented as a particle, is propagated through the river network based on a travel time (the celerity divided by the segment length). Note that the wave celerity differs from the flow velocity, as the wave typically moves faster than water mass
  - The wave celerity [LT^{-1}] is a function of channel width [L], Manning's coefficient [–], channel slope [unitless], and discharge [L^3T^{-1}].

Saint-Venant Equation

1-D conservation equation for continuity

$$\frac{\partial q}{\partial x} + \frac{\partial A}{\partial t} = 0$$

1-D conservation equation for momentum

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x} + g\frac{\partial y}{\partial x} - g(S_0 - S_f) = 0$$

a b c d e

- (a) is the local acceleration term
- (b) is the convective acceleration term
- (c) is the pressure gradient term
- (d) is the gravity term
- (e) is the friction term.

Simplification of Saint-Venant Equation in IRF

- IRF
  - **IRF** uses a **diffusive wave approximation**, which retains pressure gradient and dispersion, but neglects inertia.

$$\cancel{\frac{\partial v}{\partial t}} + v\cancel{\frac{\partial v}{\partial x}} + g\frac{\partial y}{\partial x} - g(S_0 - S_f) = 0$$

$Continuity\ Equation$

$Manning's\ Equation$

$$\frac{\partial q}{\partial t} = D\frac{\partial^2 q}{\partial x^2} - C\frac{\partial q}{\partial x}$$

Think of it as a **response kernel**—it tells you how a unit input of runoff at a grid cell spreads out over time as it travels downstream. It's derived analytically and used in a convolution framework.

It is important to note that both diffusivity (D) and wave speed (C) are constant. This is a linear advection-diffusion equation, which can be solved using ***Green's functions*** to derive the Impulse Response Function.

Simplification of Saint-Venant Equation in KWT

- KWT
  - **KWT** uses a **kinematic wave approximation**, which neglects both inertia and pressure gradient, assuming flow is driven purely by gravity and friction.

$$\cancel{\frac{\partial v}{\partial t}} + v\cancel{\frac{\partial v}{\partial x}} + g\cancel{\frac{\partial y}{\partial x}} - g(S_0 - S_f) = 0$$

$$S_0 = S_f$$

Continuity Equation

Manning's Equation

Models the **movement of a wave of water** down a slope, assuming the flow is driven purely by gravity and friction. It's more physically detailed but less flexible for large-scale or linear routing.

$$C = (\alpha + 1) \cdot (w)^{\frac{-\alpha}{\alpha+1}} \cdot \left(\frac{k}{n}\sqrt{S_0}\right)^{\frac{1}{\alpha+1}} \cdot q^{\frac{\alpha}{\alpha+1}}$$

It is important to note that the calculated wave speed (C) is NOT constant. The wave speed is a function of channel width, Manning's coefficient, channel slope, and discharge.

IRF versus KWT

Feature	Impulse Response Function (IRF)	Kinematic Wave Tracking (KWT)
Approach	Linear convolution using precomputed response functions	Solves simplified Saint-Venant equations

IRF versus KWT

Feature	Impulse Response Function (IRF)	Kinematic Wave Tracking (KWT)
Approach	Linear convolution using precomputed response functions	Solves simplified Saint-Venant equations
Flow Representation	Assumes linear, time-invariant response to runoff	Assumes gravity-driven flow with friction, neglecting pressure and inertia

IRF versus KWT

Feature	Impulse Response Function (IRF)	Kinematic Wave Tracking (KWT)
Approach	Linear convolution using precomputed response functions	Solves simplified Saint-Venant equations
Flow Representation	Assumes linear, time-invariant response to runoff	Assumes gravity-driven flow with friction, neglecting pressure and inertia
Computational Efficiency	Very efficient; IRFs are preprocessed	Moderate; requires solving PDEs over time

IRF versus KWT

Feature	Impulse Response Function (IRF)	Kinematic Wave Tracking (KWT)
Approach	Linear convolution using precomputed response functions	Solves simplified Saint-Venant equations
Flow Representation	Assumes linear, time-invariant response to runoff	Assumes gravity-driven flow with friction, neglecting pressure and inertia
Computational Efficiency	Very efficient; IRFs are preprocessed	Moderate; requires solving PDEs over time
Hydrologic Assumptions	No dynamic feedback; routing is independent of flow conditions	Assumes steady, one-dimensional flow; no backwater effects

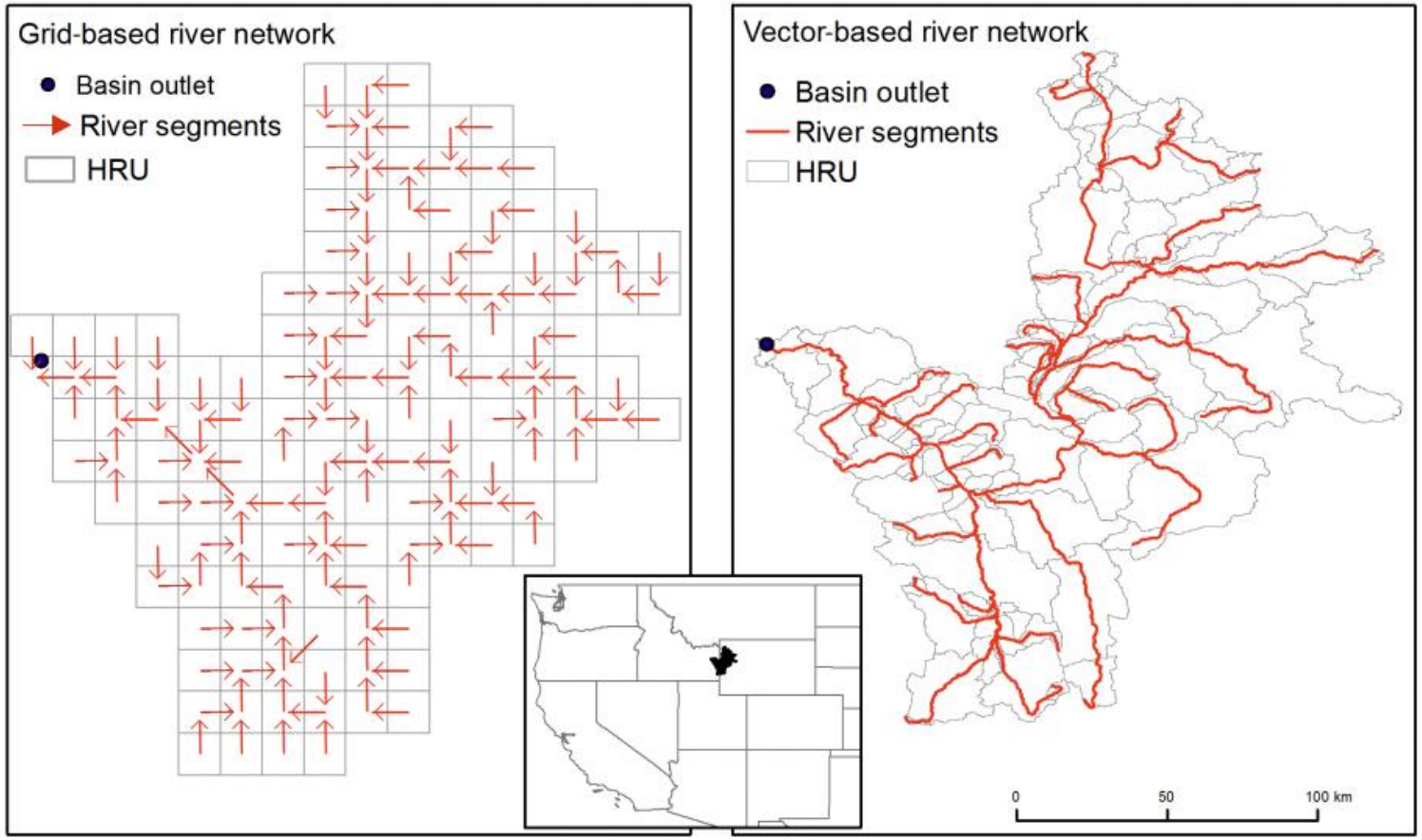
IRF versus KWT

Feature	Impulse Response Function (IRF)	Kinematic Wave Tracking (KWT)
Approach	Linear convolution using precomputed response functions	Solves simplified Saint-Venant equations
Flow Representation	Assumes linear, time-invariant response to runoff	Assumes gravity-driven flow with friction, neglecting pressure and inertia
Computational Efficiency	Very efficient; IRFs are preprocessed	Moderate; requires solving PDEs over time
Hydrologic Assumptions	No dynamic feedback; routing is independent of flow conditions	Assumes steady, one-dimensional flow; no backwater effects
Limitations	Cannot simulate dynamic channel interactions or flow reversals	Inaccurate in flat terrain or complex hydraulic conditions

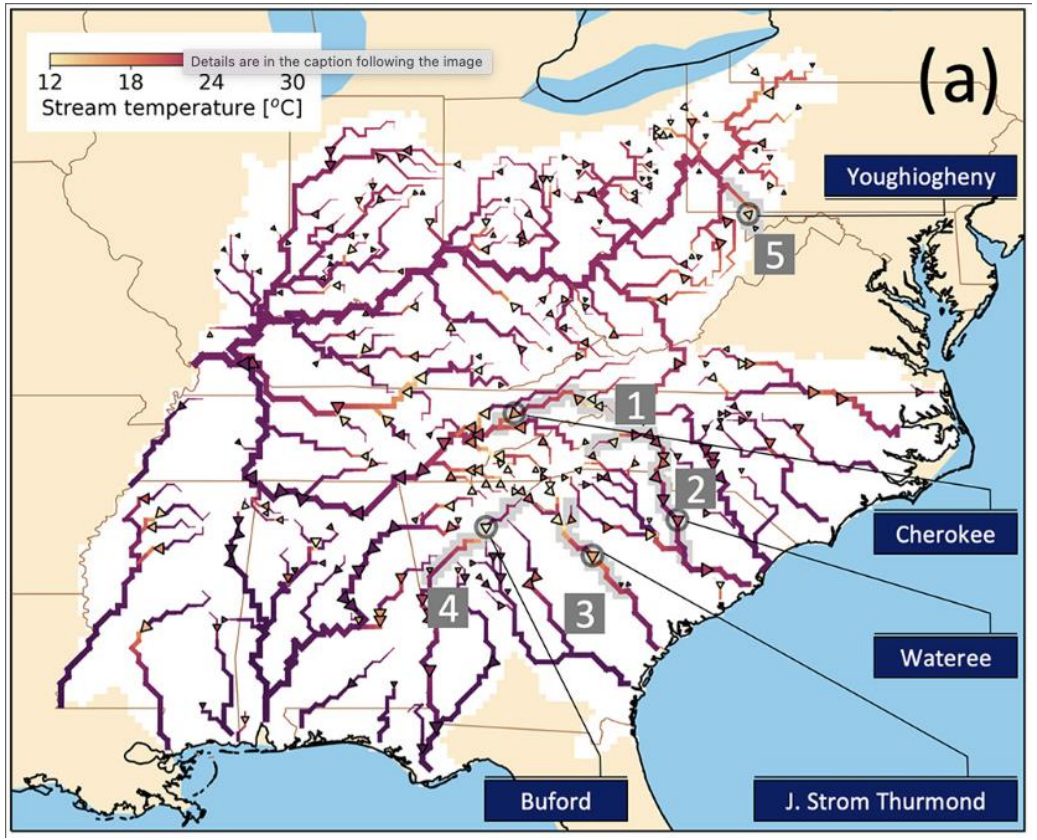
IRF versus KWT

Feature	Impulse Response Function (IRF)	Kinematic Wave Tracking (KWT)
Approach	Linear convolution using precomputed response functions	Solves simplified Saint-Venant equations
Flow Representation	Assumes linear, time-invariant response to runoff	Assumes gravity-driven flow with friction, neglecting pressure and inertia
Computational Efficiency	Very efficient; IRFs are preprocessed	Moderate; requires solving PDEs over time
Hydrologic Assumptions	No dynamic feedback; routing is independent of flow conditions	Assumes steady, one-dimensional flow; no backwater effects
Limitations	Cannot simulate dynamic channel interactions or flow reversals	Inaccurate in flat terrain or complex hydraulic conditions
Data Requirements	Flow direction, velocity, and diffusion maps	Channel geometry, slope, roughness, and inflow hydrographs

Grid-based v.s. vector-based river networks



A vector-based river network for a large river basin



The left figure shows the mean summer river temperature for major river basins in the Southeastern United States.

What are the pros/cons of a gridded river network?

Grid-based *versus* vector-based river networks

Feature	Gridded River Network	Vector-Based River Network
Implementation Difficulty	Easier to implement; Naturally following by hydrologic modeling configured in grid cells	More complex; requires detailed preprocessing

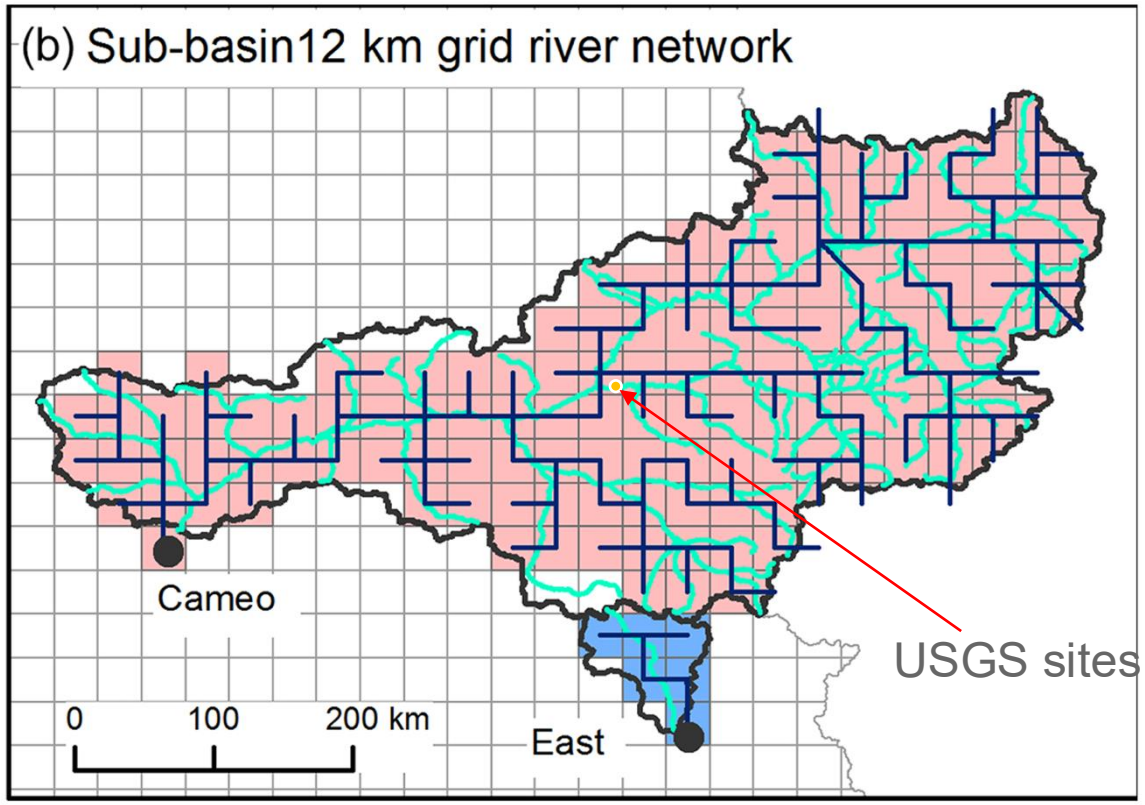
Grid-based *versus* vector-based river networks

Feature	Gridded River Network	Vector-Based River Network
Implementation Difficulty	Easier to implement; Naturally following by hydrologic modeling configured in grid cells	More complex; requires detailed preprocessing
Routing Accuracy	May misrepresent flow paths, especially in flat or complex terrain	Preserves true river geometry and connectivity

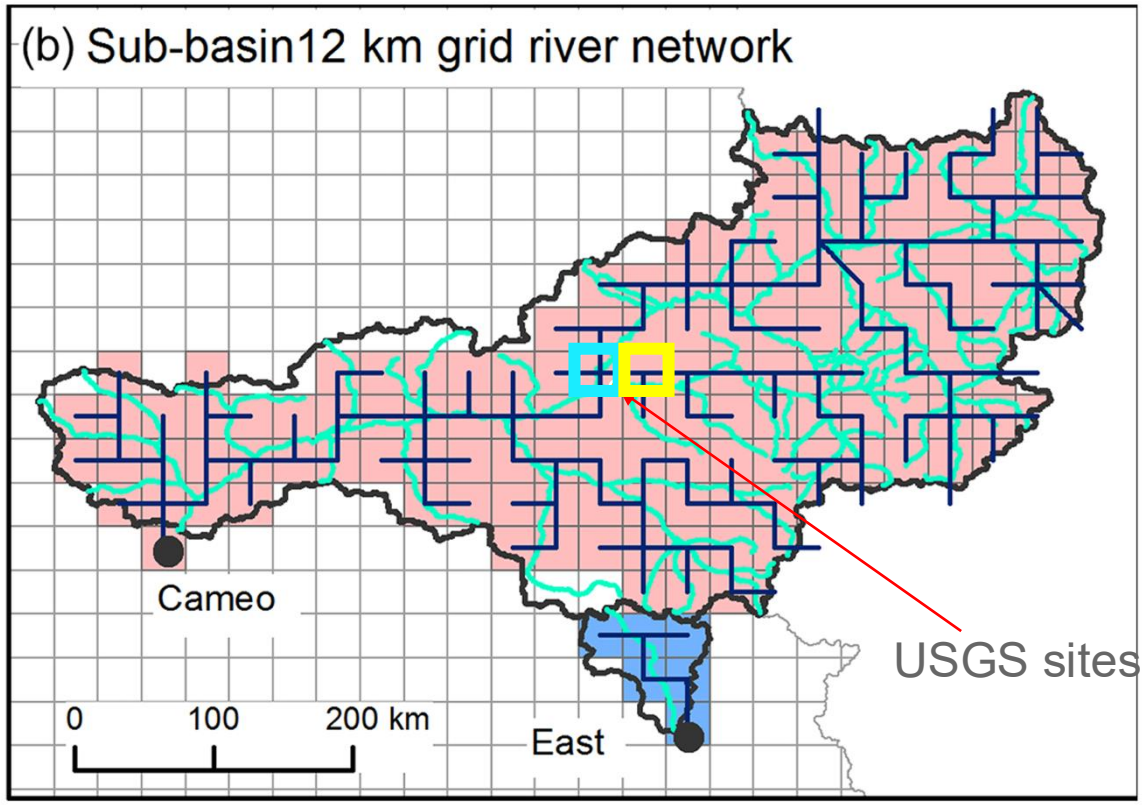
Grid-based *versus* vector-based river networks

Feature	Gridded River Network	Vector-Based River Network
Implementation Difficulty	Easier to implement; Naturally following by hydrologic modeling configured in grid cells	More complex; requires detailed preprocessing
Routing Accuracy	May misrepresent flow paths, especially in flat or complex terrain	Preserves true river geometry and connectivity
Model Evaluation	The observation sites that will be used for evaluation might not be located in the nearest grid cell.	It is very easy to find the corresponding HUCs for the observation sites.

Example: if there is a USGS site at the red dot, which grid cells of streamflow should we use to evaluate?



Example: if there is a USGS site at the red dot, which grid cells of streamflow should we use to evaluate?



Even though the USGS site is located in the grid cell with the blue boundary, we should use the yellow grid instead.

Why? The river flow in the blue grid also confluences the streamflow from the upper tributary, while the USGS site was located upstream of the confluence point.

mizuRoute – routing model

- Why name it mizuRoute?
  - ***mizu*** (水) means water in Japanese
  - The developer's last name is ***mizu***kami.
- Why mizuRoute?
  - It can be applied to a user-defined catchment-based river network (both vector-based and grid-based).
  - We can add Lakes into the river network, allowing the model to simulate discharge and volume in rivers and lakes (natural lakes or reservoirs).

What input files do we need to run mizuRoute for a vector-based river network?

Runoff data
Model output from hydrologic models

River network and topology file
River segment connection, channel slope, segment length, etc.

Parameter file
Such as C(wave velocity) and D (Diffusivity) in IRF

Mapping file
It is used to map the runoff from the grid-based network to each hydrologic response unit (HRU)

Questions?

